

Qualification Pack



Hydrogen Plant Technician (Installation, Commissioning and Maintenance)

QP Code: PSS/Q4601

Version: 1.0

NSQF Level: 4

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PSS/Q4601: Hydrogen Plant Technician (Installation, Commissioning and Maintenance)

Brief Job Description

The incumbent in the job is responsible for installation, testing, dismantling, repairing and replacing a range of instruments & faulty components, calibration & maintenance of Hydrogen Plant Instrumentation & Control System. He/She also undertakes both planned & Breakdown maintenance, setting up test equipment, troubleshooting of instruments, calibration of instruments, installation for expansion, modification & project work across the range of Instrumentation & Control System.

Personal Attributes

The job holder must have an eye for detail as well as the patience and discipline required to carry out detailed and repetitive tasks. The candidate should be able to read and understand SLD (Single line drawing, Instrumentation loop drawing, Equipment technical manuals, instructions and warnings.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N1336: Working effectively with others](#)
2. [PSS/N1337: Apply Basic Health and Safety Practices for Power Related Work](#)
3. [PSS/N4601: Carryout Installation, Testing and Calibration of Hydrogen Plant Instrumentation and Control System](#)
4. [PSS/N4605: Carry out maintenance of hydrogen plant instrumentation and control system](#)
5. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Green Hydrogen
Occupation	Green Hydrogen - Operation & Maintenance
Country	India
NSQF Level	4



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Credits	15
Aligned to NCO/ISCO/ISIC Code	NCO-2015/NIL
Minimum Educational Qualification & Experience	10th Class (pass plus completed 2nd year of 03 years Engineering Diploma in Electrical/Electronics/Instrumentation Engineering) OR 12th grade Pass (Physics Chemistry Mathematics (PCM) or Equivalent) OR 10th grade pass plus 2-year NTC (/NAC in relevant Trade) OR Previous relevant Qualification of NSQF Level (3 with minimum education as 8th Grade pass) with 3 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level with 1.5 years of experience relevant experience
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	N/A
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	22/10/2027
NSQC Approval Date	22/10/2024
Version	1.0
Reference code on NQR	QG-04-PW-03318-2024-V1-PSSC
NQR Version	1.0



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PSS/N1336: Working effectively with others

Description

This unit covers basic practices that improve effectiveness of working with others in an organizational set-up

Scope

The scope covers the following :

- Work effectively with others
- Respect diversity

Elements and Performance Criteria

Work effectively with others

To be competent, the user/individual on the job must be able to:

- PC1.** accept information and instructions from the supervisor and fellow workers and seek clarification when required
- PC2.** pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt
- PC3.** help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency
- PC4.** use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism
- PC5.** listen attentively and respond appropriately while interacting with others at work
- PC6.** act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)
- PC7.** escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict

Respect diversity

To be competent, the user/individual on the job must be able to:

- PC8.** respect gender diversity in the workplace
- PC9.** use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive
- PC10.** communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture
- PC11.** recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** key elements of active listening and assertive communication
- KU6.** importance of teamwork in organizational and individual success
- KU7.** ethics and disciplined behavior for professional success
- KU8.** expressing and addressing grievances appropriately and managing interpersonal conflict effectively
- KU9.** harassment and discrimination based on gender, disability, caste, religion and culture and how to recognize it.
- KU10.** organizational policy/ies for harassment and discrimination

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write clear and legible notes to self, colleagues and seniors to pass messages, keep records, prepare to-do lists, take down instructions
- GS2.** read basic terms and terminologies to accurately interpret work related documents, labels, supervisor instructions in the local language
- GS3.** interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the product, production plans and other associated requirements
- GS4.** display active listening skills while interacting with co-workers and other in the workplace
- GS5.** follow organization's rule-based decision-making process
- GS6.** use appropriate planning to maintain a smooth relationship with fellow team members
- GS7.** meet customer requirements
- GS8.** critically evaluate operation parameters in relation to system normality

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with others</i>	18	32	-	-
PC1. accept information and instructions from the supervisor and fellow workers and seek clarification when required	2	4	-	-
PC2. pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt	1	4	-	-
PC3. help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency	2	4	-	-
PC4. use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism	4	6	-	-
PC5. listen attentively and respond appropriately while interacting with others at work	3	6	-	-
PC6. act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)	3	5	-	-
PC7. escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict	3	3	-	-
<i>Respect diversity</i>	16	24	-	-
PC8. respect gender diversity in the workplace	4	6	-	-
PC9. use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive	4	6	-	-
PC10. communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture	4	6	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same	4	6	-	-
NOS Total	34	56	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1
Version	2.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

PSS/N1337: Apply Basic Health and Safety Practices for Power Related Work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, organizational assets and the environment.

Scope

The scope covers the following :

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)
- PC2.** identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)
- PC3.** use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment
- PC4.** follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)
- PC5.** ensure positive isolation of electrical equipment and system as per the given standards
- PC6.** identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.
- PC7.** use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)
- PC8.** use proper scaffolds, elevated platforms and ladders for working at height
- PC9.** inspect various tools and plants (T & P) routinely for any signs of oil, water leakages
- PC10.** store flammable materials and machine lubricating oil safely and correctly



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PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards

PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)

PC14. use appropriate rescue techniques during fire hazard

PC15. use good housekeeping practices at all times to avoid accidental fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)

PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real

PC18. write an incident report and send the same to the authorized person

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. names (and job titles if applicable) and location of all documents along with the people responsible for health and safety in a workplace

KU2. possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity (such as drunkenness), health hazards (such as untreated injuries and contagious illness), not taking safety precautions

KU3. safe working practices when working with tools, machines

KU4. positive isolation of electrical equipment and system

KU5. use of emission and pollution control devices and measures taken to control pollution

KU6. preventative and remedial actions to be taken in the case of exposure to toxic materials solvents, flux, lead etc.

KU7. precautionary measures taken to prevent fire accident

KU8. various causes of fire (Causes of fires: heating of metal, spontaneous ignition, sparking, electrical heating, loose fires (smoking, welding, etc.), chemical fires, etc.)

KU9. different materials (sand, water, foam, CO₂, dry powder) used in fire extinguisher and proper method of using it to extinguish fire

KU10. various types of safety signs with their names

KU11. basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries basic techniques of bandaging

KU12. artificial respiration and the CPR Process



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read & write an accident/ incident report in local language and/or English
- GS2.** read and comprehend basic content to read labels, charts, signages, manuals etc.
- GS3.** question co-workers appropriately and give clear instruction to them and sub ordinates in order to clarify instructions
- GS4.** remain congenial while discussing and debating issues with co-workers
- GS5.** observe appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS6.** plan and organize work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS7.** build customer relationships and use customer centric approach
- GS8.** seek assistance from other sources to resolve problem and report problems that cannot be resolved to the appropriate authority

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow workplace health and safety practices</i>	25	36	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)	3	5	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)	3	2	-	-
PC3. use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment	3	5	-	-
PC4. follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)	2	2	-	-
PC5. ensure positive isolation of electrical equipment and system as per the given standards	2	4	-	-
PC6. identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.	2	4	-	-
PC7. use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)	2	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. use proper scaffolds, elevated platforms and ladders for working at height	2	3	-	-
PC9. inspect various tools and plants (T & P) routinely for any signs of oil, water leakages	2	3	-	-
PC10. store flammable materials and machine lubricating oil safely and correctly	2	2	-	-
PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	3	-	-
PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	3	-	-
<i>Follow fire safety practices</i>	6	12	-	-
PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)	3	5	-	-
PC14. use appropriate rescue techniques during fire hazard	2	5	-	-
PC15. use good housekeeping practices at all times to avoid accidental fire	1	2	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	8	13	-	-
PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)	3	6	-	-
PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real	2	5	-	-
PC18. write an incident report and send the same to the authorized person	3	2	-	-
NOS Total	39	61	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1337
NOS Name	Apply Basic Health and Safety Practices for Power Related Work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



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PSS/N4601: Carryout Installation, Testing and Calibration of Hydrogen Plant Instrumentation and Control System

Description

This unit is about installation, testing and calibration of green hydrogen plant instrumentation and control system under the guidance of supervisor/engineer and as per the specified standards and procedures.

Scope

The scope covers the following :

- Preparing for the installation, testing & calibration activities
- Installation, testing & calibration of instrumentation & control system
- Perform post-installation, testing & calibration activities

Elements and Performance Criteria

Preparing for the installation activities

To be competent, the user/individual on the job must be able to:

- PC1.** Read and interpret information received from instrument documents P & I diagram (process & instrumentation diagram), instrument index/data sheets/hook up diagram/loop diagram/instrument layout diagram/wiring diagrams etc.) and instructions received from the supervisor/engineer related to installation of instrumentation & control system in green hydrogen plant
- PC2.** Obtain work permit from the process supervisor/engineer/authority before commencing work
- PC3.** Carry out a site check including visually prior to installation activities
- PC4.** Follow all relevant drawings and specifications for the installation being carried out
- PC5.** Select & arrange appropriate tools, equipment, instruments and control system (i.e. analysers, sensors, controllers etc.) as per the plant requirement
- PC6.** Check that tools, equipment, instruments & control system are safe and functioning properly before installation
- PC7.** Carry out safe isolation of power source as per applicable specification/ electrical safety norms
- PC8.** Follow safe working practices in accordance with the organizational standards and regulations
- PC9.** Inform all affected parties of the intended work plan prior to disconnecting power supply line

Installation and testing of instrumentation and control system

To be competent, the user/individual on the job must be able to:

- PC10.** Ensure that the instruments need to install are calibrated according to specifications and properly connected to the control system
- PC11.** Mount the instruments and sensors in their designated locations
- PC12.** Apply appropriate methods to install the instruments and sensors as per the drawing and SOP



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- PC13.** Route & connect the various electrical & communication cables, and setting up the control system
- PC14.** Ensure that new electrical & instrumentation cable installed should not be left hanging and proper instrumentation cable tray to be utilized (Ferruling of the cable on both side is mandatory)
- PC15.** Ensure that new electrical & instrumentation cable installed should not be left hanging and proper instrumentation cable tray to be utilised (Ferruling of the cable on both side is mandatory)
- PC16.** Carry out trial run and perform applicable tests to ensure safe and desired working of power supply equipment
- PC17.** Conduct continuity test of signal from the instruments to the control panel i.e. Programmable Logic Controllers (PLC), Distributed Control Systems (DCS), Integrated Computer Control System (ICCS) or any other logical system
- PC18.** Test control panel signal from/to all instruments and actuators using appropriate signal generator depending on instrument type
- PC19.** Test ESD operation of plant to generate the required pressure, level or other parameter
- PC20.** Perform loop test to test the loops for different type of control action such as direct, reverse, split range etc
- PC21.** Record all data in the loop folder
- PC22.** Check that signal from the instrument is received by the control algorithm and the appropriate corrective output signal is generated and received by the final control element
- PC23.** Co-ordinate with the team working at the end with PLC/DCS/other system, as per the loop testing procedure
- PC24.** Prepare the tests report as defined in the SOP

Perform post-installation and testing activities

To be competent, the user/individual on the job must be able to:

- PC25.** Check the list of corrections i.e. adjustment, refitting, cleaning, overhaul, recalibration, wiring change, replacement etc. need to be carried out from the tests report
- PC26.** Take appropriate actions and resolve the issues under guidance of supervisor/engineer
- PC27.** Calibrate the instruments using appropriate methods and devices in line with manufacturer's quality standards and regulations
- PC28.** Prepare records and documents related to the work done
- PC29.** Ensure tagging of embedded, exposed electrical lines, its accessories and other equipment
- PC30.** Leave the work area in clean and tidy condition after the completion of work

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the organisation
- KU2.** relevant health and safety requirements applicable in the workplace
- KU3.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU4.** basic electrical concepts such as power factor, current and voltage transmission
- KU5.** various symbols and terminologies used in smart metering systems



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- KU6.** role of various elements of smart metering system such as HES, communication platform, MDMS etc.
- KU7.** networking for smart meter installation such as configuring IP address, implementation of networking topologies such as HAN (Home Area Network), WAN (Wide Area Network) and MAN (Metropolitan Area Network), establishing communication, etc.
- KU8.** basics of communication technologies such as RF, GSM/CDMA, fiber optic communications, PLCs, Zigbee, 6LoWPAN, Bluetooth etc.
- KU9.** identification of different ports used in the installation of smart meters such as RS 232, RS 485 etc.
- KU10.** importance of job specifications in the installation and commissioning process
- KU11.** considerations for selecting an appropriate location for smart meter installation
- KU12.** tools, equipment, hardware/software technologies, wiring, cabling and other materials required for smart meter installation and commissioning
- KU13.** installation of the In-Home Display (IHD) in an appropriate location
- KU14.** procedures for various checks and tests. Checks and tests: Testing the functionalities of smart meter, checking basic connections, features, switches within the smart metering network; testing of the IHD as per required standards; checking of the communication capabilities between HES and smart meter
- KU15.** procedure to provide adequate protection to the smart meter
- KU16.** problematic issues and situations that require expert intervention
- KU17.** document necessary information such as observations, work reports, key findings, etc.
- KU18.** relevant standards/regulations/prevaling practices of the utility for smart meter installation and commissioning

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Note the information related to work and processes
- GS2.** Write reports and observations related to work in English/regional language
- GS3.** Read and interpret and process flowchart for all operations
- GS4.** Read symbols, manuals and operation documents to understand the Equipment used into operation
- GS5.** Discuss task lists, schedules and activities with the seniors and team members
- GS6.** Follow organization rule-based decision making process
- GS7.** Take decisions with systematic course of actions and/or response
- GS8.** Plan and organize tasks to meet deadlines
- GS9.** Find ways of modifying difficult operating stages to make it operation friendly
- GS10.** Apply domain information to set and define operation parameters that ensures economy and quality of the product
- GS11.** Analyse the complexity of work to determine if it can be successfully carried out or needs to be referred to a superior/specialist
- GS12.** Recognise a workplace problem and take suitable action to resolve it



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Preparing for the installation activities</i>	11	10	-	-
PC1. Read and interpret information received from instrument documents P & I diagram (process & instrumentation diagram), instrument index/data sheets/hook up diagram/loop diagram/ instrument layout diagram/wiring diagrams etc.) and instructions received from the supervisor/engineer related to installation of instrumentation & control system in green hydrogen plant	1	1	-	-
PC2. Obtain work permit from the process supervisor/engineer/authority before commencing work	1	1	-	-
PC3. Carry out a site check including visually prior to installation activities	1	1	-	-
PC4. Follow all relevant drawings and specifications for the installation being carried out	1	-	-	-
PC5. Select & arrange appropriate tools, equipment, instruments and control system (i.e. analysers, sensors, controllers etc.) as per the plant requirement	3	2	-	-
PC6. Check that tools, equipment, instruments & control system are safe and functioning properly before installation	1	2	-	-
PC7. Carry out safe isolation of power source as per applicable specification/ electrical safety norms	1	1	-	-
PC8. Follow safe working practices in accordance with the organizational standards and regulations	1	1	-	-
PC9. Inform all affected parties of the intended work plan prior to disconnecting power supply line	1	1	-	-
<i>Installation and testing of instrumentation and control system</i>	22	37	-	-
PC10. Ensure that the instruments need to install are calibrated according to specifications and properly connected to the control system	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. Mount the instruments and sensors in their designated locations	2	3	-	-
PC12. Apply appropriate methods to install the instruments and sensors as per the drawing and SOP	3	4	-	-
PC13. Route & connect the various electrical & communication cables, and setting up the control system	2	3	-	-
PC14. Ensure that new electrical & instrumentation cable installed should not be left hanging and proper instrumentation cable tray to be utilized (Ferruling of the cable on both side is mandatory)	1	2	-	-
PC15. Ensure that new electrical & instrumentation cable installed should not be left hanging and proper instrumentation cable tray to be utilised (Ferruling of the cable on both side is mandatory)	2	3	-	-
PC16. Carry out trial run and perform applicable tests to ensure safe and desired working of power supply equipment	1	2	-	-
PC17. Conduct continuity test of signal from the instruments to the control panel i.e. Programmable Logic Controllers (PLC), Distributed Control Systems (DCS), Integrated Computer Control System (ICCS) or any other logical system	3	5	-	-
PC18. Test control panel signal from/to all instruments and actuators using appropriate signal generator depending on instrument type	1	2	-	-
PC19. Test ESD operation of plant to generate the required pressure, level or other parameter	1	2	-	-
PC20. Perform loop test to test the loops for different type of control action such as direct, reverse, split range etc	1	2	-	-
PC21. Record all data in the loop folder	1	1	-	-
PC22. Check that signal from the instrument is received by the control algorithm and the appropriate corrective output signal is generated and received by the final control element	1	2	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. Co-ordinate with the team working at the end with PLC/DCS/other system, as per the loop testing procedure	1	2	-	-
PC24. Prepare the tests report as defined in the SOP	1	2	-	-
<i>Perform post-installation and testing activities</i>	7	13	-	-
PC25. Check the list of corrections i.e. adjustment, refitting, cleaning, overhaul, recalibration, wiring change, replacement etc. need to be carried out from the tests report	2	4	-	-
PC26. Take appropriate actions and resolve the issues under guidance of supervisor/engineer	1	2	-	-
PC27. Calibrate the instruments using appropriate methods and devices in line with manufacturer's quality standards and regulations	1	2	-	-
PC28. Prepare records and documents related to the work done	1	2	-	-
PC29. Ensure tagging of embedded, exposed electrical lines, its accessories and other equipment	1	2	-	-
PC30. Leave the work area in clean and tidy condition after the completion of work	1	1	-	-
NOS Total	40	60	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N4601
NOS Name	Carryout Installation, Testing and Calibration of Hydrogen Plant Instrumentation and Control System
Sector	Power
Sub-Sector	Distribution
Occupation	Green Hydrogen - Operation & Maintenance
NSQF Level	4
Credits	6
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

PSS/N4605: Carry out maintenance of hydrogen plant instrumentation and control system

Description

This unit is about performing various tasks for Breakdown, (Repair , Replacement) , routine and preventive maintenance of hydrogen plant instrumentation and control system.

Scope

The scope covers the following :

- Prepare for breakdown, routine and preventive maintenance work
- Perform breakdown, routine and preventive maintenance activities
- Perform post-maintenance activities

Elements and Performance Criteria

Prepare for breakdown, routine and preventive maintenance work

To be competent, the user/individual on the job must be able to:

- PC1.** Determine the routine maintenance requirements of instrumentation & control system by interpreting manufacturer's manual, maintenance checklist & schedule, work instructions, etc.
- PC2.** Read maintenance schedule and checklist and plan the time and schedule for conducting the maintenance activities
- PC3.** Arrange tools, consumables, spare parts, etc. required for performing repair & maintenance
- PC4.** Clear the work area thoroughly to ensure no unwanted materials are present before starting the work
- PC5.** Ensure that instrumentation and control system is not connected with the equipment or any power source before starting maintenance work
- PC6.** Adhere to all safety norms (such as wearing protective gloves, shoes, etc.) in accordance with the international/ national and organisational standards
- PC7.** Check for ESD symbol on electronic instrumentation and accordingly use ESD protection devices to avoid damage
- PC8.** Inform all affected parties including authorities/ process supervisors of the intended work plan prior to disconnecting power supply line

Perform breakdown, routine and preventive maintenance activities

To be competent, the user/individual on the job must be able to:

- PC9.** Perform basic check-up of instruments, sensors, controllers etc. as specified in the manufacturer manual
- PC10.** Carry out the maintenance activities on a range of instrumentation and control equipment, following the appropriate techniques and procedures
- PC11.** Check the technical and mechanical systems to ensure their operation within the specified parameters



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- PC12.** Conduct instrumentation evaluation for hydraulic, pneumatic, mechanical, electrical and electronic measuring & control equipment
- PC13.** Identify the type of faults, required tools and devices for removal of faults
- PC14.** Isolate the faulty part and repair or replace it as per requirement and by following correct procedures
- PC15.** Dismantle the equipment and replace/change or rectify the instrumentation / electrical system
- PC16.** Use spare parts and consumables of equipment as per manufacturer specification
- PC17.** Check the internal conditions of wiring and other parts etc. to test its working status and expected conditions
- PC18.** Conduct breakdown maintenance and then inspect the equipment for leaks, short circuit in parts, breakage of wires and clamps, vibrations, erratic/problematic performance or any other problem as reported in the complain
- PC19.** Conduct the appropriate tests related to the functional performance of instruments, as required
- PC20.** Carry out troubleshooting for test failures to determine root causes and develop processes to avoid repeat failure
- PC21.** Use the appropriate scientific and analytic computer software, calculators, electronic probes, voltage and current meters to diagnose faults in circuitry
- PC22.** Coordinate with the Process supervisor / relevant personnel to evaluate the functioning of instruments and equipment
- PC23.** Use the relevant testing devices for instrumentation to measure & control pressure, flow, temperature, level, motion, force, and chemical makeup
- PC24.** Perform adjustment/ calibration/part replacement/ instrument replacement as specified
- PC25.** Coordinate with Process engineers or process technicians to resolve the issues requiring their assistance
- PC26.** Recommission the instrumentation system appropriately, to return it to service, following repair & replacement activities
- PC27.** Clean & tighten the terminals, screws, clamps and cables and ensure that they are free from dirt, filth, soot and corrosion by following the instructions given in manufacturer manual
- PC28.** Assess when the problem is beyond Technician's competence and report it to supervisor/ relevant authority

Perform post-maintenance activities

To be competent, the user/individual on the job must be able to:

- PC29.** Assemble back the equipment and prepare it for conducting the trials
- PC30.** Leave the work area in clean and tidy condition after the completion of work
- PC31.** Dispose off waste materials such as failed parts/aggregates, as per organisation's and environmental policies
- PC32.** Record / document all repairs carried out, time taken and other significant findings observed during the work process
- PC33.** Ensure that all maintenance activities are adequately addressed

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** Relevant legislation, standards, policies, and procedures followed in the organization
- KU2.** The applicable isolation and lock-off procedures or permit-to-work procedure that applies
- KU3.** How to interpret drawings, specifications, manufacturers manuals and other documents needed during the maintenance process
- KU4.** Functioning of different process plants and the appropriate measuring and control equipment
- KU5.** Process to be adopted to establish the background of the fault
- KU6.** How to evaluate various types of information available for fault diagnoses such as operator-provided information, equipment self-diagnosis, and testing instrument measurements
- KU7.** Use of various testing instruments such as, multi-meter, oscilloscope, logic probe, signal tracer, signal generator, etc.
- KU8.** Use of various aids for fault diagnosis such as logic diagrams fault analysis charts, flow charts or algorithms, manufacturers' manuals, probability charts, computer-aided test equipment, electronic aids, etc.
- KU9.** The application of various fault-finding techniques such as half-split technique, input/output technique, injection and sampling, six-point technique, emergent sequence, unit substitution, function/performance testing equipment self-diagnostics, etc.
- KU10.** How to evaluate sensory conditions i.e. sight, sound, smell, touch
- KU11.** How to analyze evidence and evaluate possible characteristics and causes of specific faults/problems
- KU12.** How to relate to the previous reports/records of similar fault conditions
- KU13.** How to evaluate the likely risk of running the equipment with the displayed fault, and the effects the fault could have on health and safety, and on the overall process or system
- KU14.** The process of handling and maintaining instrumentation test instruments
- KU15.** How to check that test instruments are within current calibration dates, and that they are free from damage and defects
- KU16.** Relevant precautions to be taken to prevent Electrostatic Discharge (ESD) damage to electronic circuits and components
- KU17.** Basic principles of operation of the instrumentation and control equipment
- KU18.** How the instrumentation system functions and its operating sequence
- KU19.** Working purpose of individual units/components such as gauges and meters and how they interact
- KU20.** Importance of ensuring the control systems are isolated or put into manual control, and appropriate trip locks, keys or program overrides are inserted before removing any sensors or instruments from the system
- KU21.** Process of identifying and selecting instrument sensors, including how to identify their markings calibration information component values operating parameters and working range
- KU22.** Appropriate way of fitting instruments to avoid faulty readings caused by head correction, poor flow past sensor, blockages, incorrect wiring, poor insulation or incorrect materials
- KU23.** How to carry out visual checks of the instruments including checking for leaks, security of joints and physical damage
- KU24.** Relevant replacement parts, materials and other items required for the maintenance process



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- KU25.** Relevant techniques used to dismantle/assemble integrated equipment i.e. release of pressures/force, proof marking to aid reassembly, plugging exposed pipe/component openings, dealing with soldered joints, screwed, clamped and crimped connections
- KU26.** Importance and process of attaching identification marks/labels to removed components or cables, to assist with reassembly
- KU27.** Applicable methods for checking components are fit for purpose, and the need to replace electronic modules, sensors, transmitters, transducers, electronic boards and other failed items
- KU28.** How to resolve the common problems encountered during the maintenance of the instrumentation and control system
- KU29.** The importance of making process improvements by engaging the relevant stakeholders
- KU30.** How to create and update the SOPs, maintenance schedules and plans
- KU31.** Process of carrying out different types of maintenance such as preventive, corrective, predictive and reactive maintenance, and the activities these entail

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Note the information related to work and processes
- GS2.** Write reports and observations related to work in English/regional language
- GS3.** Read and interpret and process flowchart for all operations
- GS4.** Read symbols, manuals and operation documents to understand the Equipment used into operation
- GS5.** Discuss task lists, schedules and activities with the seniors and team members
- GS6.** Follow organization rule-based decision making process
- GS7.** Take decisions with systematic course of actions and/or response
- GS8.** Plan and organize tasks to meet deadlines
- GS9.** Find ways of modifying difficult operating stages to make it operation friendly
- GS10.** Apply domain information to set and define operation parameters that ensures economy and quality of the product
- GS11.** Analyse the complexity of work to determine if it can be successfully carried out or needs to be referred to a superior/specialist
- GS12.** Recognise a workplace problem and take suitable action to resolve it

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for breakdown, routine and preventive maintenance work</i>	10	10	-	-
PC1. Determine the routine maintenance requirements of instrumentation & control system by interpreting manufacturer's manual, maintenance checklist & schedule, work instructions, etc.	1	1	-	-
PC2. Read maintenance schedule and checklist and plan the time and schedule for conducting the maintenance activities	1	1	-	-
PC3. Arrange tools, consumables, spare parts, etc. required for performing repair & maintenance	3	2	-	-
PC4. Clear the work area thoroughly to ensure no unwanted materials are present before starting the work	1	1	-	-
PC5. Ensure that instrumentation and control system is not connected with the equipment or any power source before starting maintenance work	1	1	-	-
PC6. Adhere to all safety norms (such as wearing protective gloves, shoes, etc.) in accordance with the international/ national and organisational standards	1	1	-	-
PC7. Check for ESD symbol on electronic instrumentation and accordingly use ESD protection devices to avoid damage	1	2	-	-
PC8. Inform all affected parties including authorities/ process supervisors of the intended work plan prior to disconnecting power supply line	1	1	-	-
<i>Perform breakdown, routine and preventive maintenance activities</i>	25	43	-	-
PC9. Perform basic check-up of instruments, sensors, controllers etc. as specified in the manufacturer manual	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. Carry out the maintenance activities on a range of instrumentation and control equipment, following the appropriate techniques and procedures	2	3	-	-
PC11. Check the technical and mechanical systems to ensure their operation within the specified parameters	2	3	-	-
PC12. Conduct instrumentation evaluation for hydraulic, pneumatic, mechanical, electrical and electronic measuring & control equipment	2	3	-	-
PC13. Identify the type of faults, required tools and devices for removal of faults	1	2	-	-
PC14. Isolate the faulty part and repair or replace it as per requirement and by following correct procedures	1	2	-	-
PC15. Dismantle the equipment and replace/change or rectify the instrumentation / electrical system	1	2	-	-
PC16. Use spare parts and consumables of equipment as per manufacturer specification	1	1	-	-
PC17. Check the internal conditions of wiring and other parts etc. to test its working status and expected conditions	1	2	-	-
PC18. Conduct breakdown maintenance and then inspect the equipment for leaks, short circuit in parts, breakage of wires and clamps, vibrations, erratic/problematic performance or any other problem as reported in the complain	2	4	-	-
PC19. Conduct the appropriate tests related to the functional performance of instruments, as required	1	2	-	-
PC20. Carry out troubleshooting for test failures to determine root causes and develop processes to avoid repeat failure	2	4	-	-
PC21. Use the appropriate scientific and analytic computer software, calculators, electronic probes, voltage and current meters to diagnose faults in circuitry	1	2	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. Coordinate with the Process supervisor / relevant personnel to evaluate the functioning of instruments and equipment	1	2	-	-
PC23. Use the relevant testing devices for instrumentation to measure & control pressure, flow, temperature, level, motion, force, and chemical makeup	1	2	-	-
PC24. Perform adjustment/ calibration/part replacement/ instrument replacement as specified	1	2	-	-
PC25. Coordinate with Process engineers or process technicians to resolve the issues requiring their assistance	1	1	-	-
PC26. Recommission the instrumentation system appropriately, to return it to service, following repair & replacement activities	1	1	-	-
PC27. Clean & tighten the terminals, screws, clamps and cables and ensure that they are free from dirt, filth, soot and corrosion by following the instructions given in manufacturer manual	1	2	-	-
PC28. Assess when the problem is beyond Technician's competence and report it to supervisor/ relevant authority	1	1	-	-
<i>Perform post-maintenance activities</i>	5	7	-	-
PC29. Assemble back the equipment and prepare it for conducting the trials	1	2	-	-
PC30. Leave the work area in clean and tidy condition after the completion of work	1	1	-	-
PC31. Dispose off waste materials such as failed parts/aggregates, as per organisation's and environmental policies	1	1	-	-
PC32. Record / document all repairs carried out, time taken and other significant findings observed during the work process	1	2	-	-
PC33. Ensure that all maintenance activities are adequately addressed	1	1	-	-



Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	40	60	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	PSS/N4605
NOS Name	Carry out maintenance of hydrogen plant instrumentation and control system
Sector	Power
Sub-Sector	Distribution
Occupation	Green Hydrogen - Operation & Maintenance
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team



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Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services



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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitation to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2029
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass 70% aggregate for the QP.



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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1336.Working effectively with others	34	56	-	-	90	10
PSS/N1337.Apply Basic Health and Safety Practices for Power Related Work	39	61	-	-	100	10
PSS/N4601.Carryout Installation, Testing and Calibration of Hydrogen Plant Instrumentation and Control System	40	60	-	-	100	35
PSS/N4605.Carry out maintenance of hydrogen plant instrumentation and control system	40	60	-	-	100	35
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	10
Total	173	267	-	-	440	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.